Introduction to Artificial Intelligence

Intelligent Systems — Basic Concept; Intelligent Agents and Rational Agents

Intelligent Systems — Basic Concept

An intelligent system is a machine, software, or a combination of hardware and software that can:

1. Perceive its surroundings or receive data. ➔ Input
2. Process the information it receives. ➔Processing
3. Make decisions based on that information. ➔ Decision
4. Take action to achieve a goal. ➔ Action
5. Learn or improve from experience, if it has a learning capability. ➔ Feedback

Core ingredients of an intelligent system

- **Input / perception:** sensors, data feeds, user inputs.
- **Knowledge / internal state:** rules, models, learned parameters, memory.
- **Reasoning / decision-making:** choosing what to do next.
- **Output / action:** displaying a result, moving a motor, sending a message.
- **Feedback / learning:** using the outcome of actions to improve.

Examples: Smart AC , a spam filter, a thermostat that learns your schedule.

Intelligent Agents

An agent is anything that perceives its environment through sensors and acts upon that environment through actuators. The word "intelligent" is added when the agent's actions are chosen to achieve good outcomes.

Agent	Sensors (Input Receiver)	Actuators (Action Performer)
Human	Eyes, ears, skin, nose	Hands, legs, voice
Robot vacuum	Bump sensors, cliff sensors, camera	Wheels, brushes, suction motor
Software agent (spam filter)	Incoming email text and metadata	Move to spam folder, flag, deliver
Self-driving car	Cameras, LiDAR, radar, GPS	Steering, throttle, brakes

Properties of an intelligent agent

- Autonomy: operates without constant human control.
- Reactivity: responds in time to changes in the environment. (E.g. Autonomous car brake applying)
- Proactiveness: takes initiative to pursue goals, not only react.
- Social ability: can interact or cooperate with other agents or people.

Rational Agents

A rational agent selects the action that is expected to maximize its performance measure, based on:

- The percept sequence so far.
- Its prior knowledge.
- The actions available to it.
- The performance measure.

In simple words:
A rational agent chooses the best action it can reasonably choose using the information available to it.

What rationality depends on

- The performance measure that defines success
  Example: For a self-driving car, safety may be more important than speed.
- The agent's prior knowledge of the environment.
  Example: It knows traffic rules and road maps.
- The actions the agent can perform.
  Example: A car can brake, steer or accelerate, but it cannot fly over a traffic jam.
- The percept sequence to date.
  Example: It observed a red light, then a pedestrian, then a vehicle approaching from behind.

Rational does not mean

- **Omniscient:** a rational agent maximises expected performance, not actual performance. If you cross an empty road and a cargo door falls from a plane, you were not irrational.
- **Perfect or clairvoyant:** it acts on what it can perceive and predict.
- **Successful every time:** uncertainty can lead to bad outcomes despite good decisions.

Rationality also includes information gathering (looking before crossing) and learning from percepts.

Agent, Environment and the Agent Function; PEAS Description

Agent and Environment

The agent and its environment form a loop:

The environment presents precepts to the agent, and
The agent's actions change the environment.

Everything outside the agent that it perceives or affects is the environment.

- Percept: the agent's input at a single instant.
- Percept sequence: the complete history of everything the agent has perceived.
- Action: what the agent outputs through its actuators.

For example:

1. Environment: A square of the floor contains dust.
2. Percept: The robot detects dirt.
3. Decision: Clean this square.
4. Action: Perform Suck.
5. Environment: The square becomes clean.
6. The robot observes again.

This cycle continues.

Agent Function vs Agent Program

- **Agent function:** an abstract mathematical mapping from every possible percept sequence to an action:
  $f : P^* \rightarrow A$. It describes what the agent does.

  The agent function tells us what action should be selected for a given percept history.

- **Agent program:** An agent program is the actual code or algorithm that implements the agent's decision-making behavior (computer + sensors + actuators). It describes how it is done.

Agent function = What the agent should do.

Agent program = How the agent implements that behavior.

Example: the two-square vacuum world

Imagine two Squares A and B, each Clean or Dirty.
The agent perceives its current location A or B and the status of current square whether that square is dirty;
Actions are Left, Right, Suck.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean] ...	Right (table continues for longer sequences)

PEAS Description

PEAS is a framework for specifying the task environment of an agent before designing it.

It helps us identify what the agent needs to do and what it needs to interact with before designing the agent.

- **P — Performance measure:** how success is judged.
- **E — Environment:** the world the agent operates in.
- **A — Actuators:** how the agent acts.
- **S — Sensors:** how the agent perceives.

Worked PEAS examples

Agent	Performance	Environment	Actuators	Sensors
Automated taxi	Safe, fast, legal, comfortable trip, profit	Roads, traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, LiDAR, GPS, speedometer, odometer
Medical diagnosis system	Healthy patient, low cost	Patient, hospital, staff	Display of questions, tests, diagnoses	Keyboard entry of symptoms, findings, answers
Part-picking robot	% of parts in correct bins	Conveyor belt with parts, bins	Jointed arm and hand	Camera, joint-angle sensors
Chatbot (customer support)	Resolution rate, user satisfaction, response time	Chat interface, users, knowledge base	Text responses, ticket creation	User text input, session history